095_100_PHY_SCI_NA_S_CH04_L2_668304.indd Page 96 06/07/19 10:48 PM f-02752/GO02887/Inspire_Science_Physical/NA/SE/Inspire_Science_Physical/G1/007_66830 ...

Different Forms of Energy

Turn on an electric light, and a dark room becomes bright. Turn on a portable music player, and sound comes through your headphones. In both situations, a change occurs. These changes differ from each other and from the tennis racket hitting the tennis ball in **Figure 6.** This is because energy has many different forms. These forms include mechanical energy, electrical energy, chemical energy, and radiant energy.

Figure 7 shows some everyday situations in which you might notice energy. Automobiles make use of the chemical energy of gasoline. Many household appliances require electrical energy to function. Radiant energy from the Sun warms Earth. In short, energy plays a role in every activity that you do.

Get It?
Identify three different forms of energy.

An energy analogy

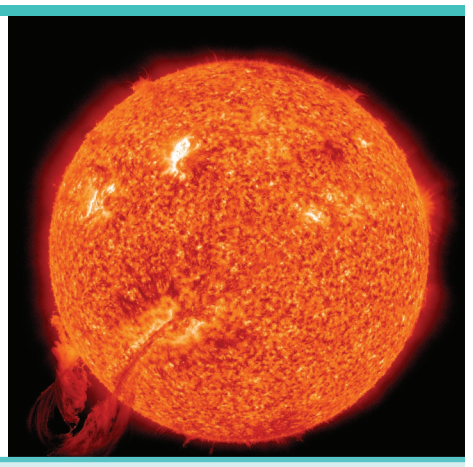
Is the chemical energy of food the same as the energy that comes from the Sun or the energy that comes from gasoline? Money can be used in an analogy to help you understand energy. Money exists in a variety of forms, such as coins, dollar bills, and twenty-dollar bills. You can convert money from one form to another. For example, you could obtain four quarters for a dollar bill. Regardless of its form, money is money. The same is true for energy. Energy from the Sun that warms you and energy from the food that you eat are only different forms of the same thing.



Chemical potential energy



Electrical energy



Radiant energy

Figure 7 Energy can be stored and transferred from one place to another. For example, the energy due to the chemical bonds in gasoline is easy to transport in cars. Electrical energy is transferred from a power plant to appliances in your home. Radiant energy is transferred from the Sun to Earth.

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SCIENCE USAGE v. COMMON USAGE

energy

Science usage: the ability to cause change
You transfer energy when you do work.

Common usage: the capacity of acting or being active
That soccer player had a lot of energy on the field today.

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